

RFI Report for August 2024

Executive summary

In general, the samples that have been collected for the Month of August 2024 appear to be relatively free from unknown RFI emissions. However, weak contaminants can be noticed in the quiet part of the band of the dataset collected on the [22nd of August 2024](#). It is also apparent from all the corresponding plots (auto, short and long baseline cross correlation) of each sample that the known RFI: GPS L1/L2/L5, Galileo 1, GPS L2, Glonass L1/L2, Galileo 2 and Inmarsat mainly affect the autocorrelation and long baseline cross correlation products. The overall RFI level for August 2024 at L-band was relatively moderate, averaging around 32% throughout the day. However, a notable spike was observed at 18:00 UTC, where the median RFI percentage across all observations reached 60%, indicating that more than half of the data at that hour was flagged as RFI

1. Introduction

The purpose of this document is to report the current status of RFI on-site. Given the construction work that has been going on over the last couple of months, it is therefore imperative to monitor the construction related activities on site that could potentially generate RFI signals which, in turn, contaminate the science observations. Preferably these reports are to be released on a weekly basis (or every two weeks).

2. Method

The flags data, specifically *ingest_flags*, are considered in these analyses, as they can be used as proxies of RFI frequency occupancy, i.e. RFI emission as a function of frequency channels. The strategy is to select two to three observations per week and report the status of the RFI contamination on site based on the flags of each dataset (of each observation). For each observation

- Autocorrelation flags are selected.
- Autocorrelation flags, which are three dimensional arrays, i.e. *time: frequency channels: correlation products*, are averaged over both time and correlation products.
- Autocorrelation flags become a one dimensional array which is plotted in order to check the RFI occupancy in the frequency channels.

The cross correlation flags are also investigated in order to see how much the RFI emission affects the correlation products which are the main interest in an interferometric observation. The procedure for the cross correlation flags is the same as that of autocorrelation ones. To further go into more detail, the flags corresponding to short and long baselines are split into two parts, and each part is plotted separately. In what follows, it is worth noting that the vertical/horizontal gray shaded areas in all plots denote the known RFI emissions in L-band.

3. Results

a. 2024, 06 August

Based on the flags of all the products of the observation run on the 6th of August 2024 (Figures 1, 2 and 3), the dataset is relatively clean. Nevertheless, there are two channels around 1500 MHz that are affected by RFI emission whose corresponding fractions of time flagged are 0.1 and ~0.2.

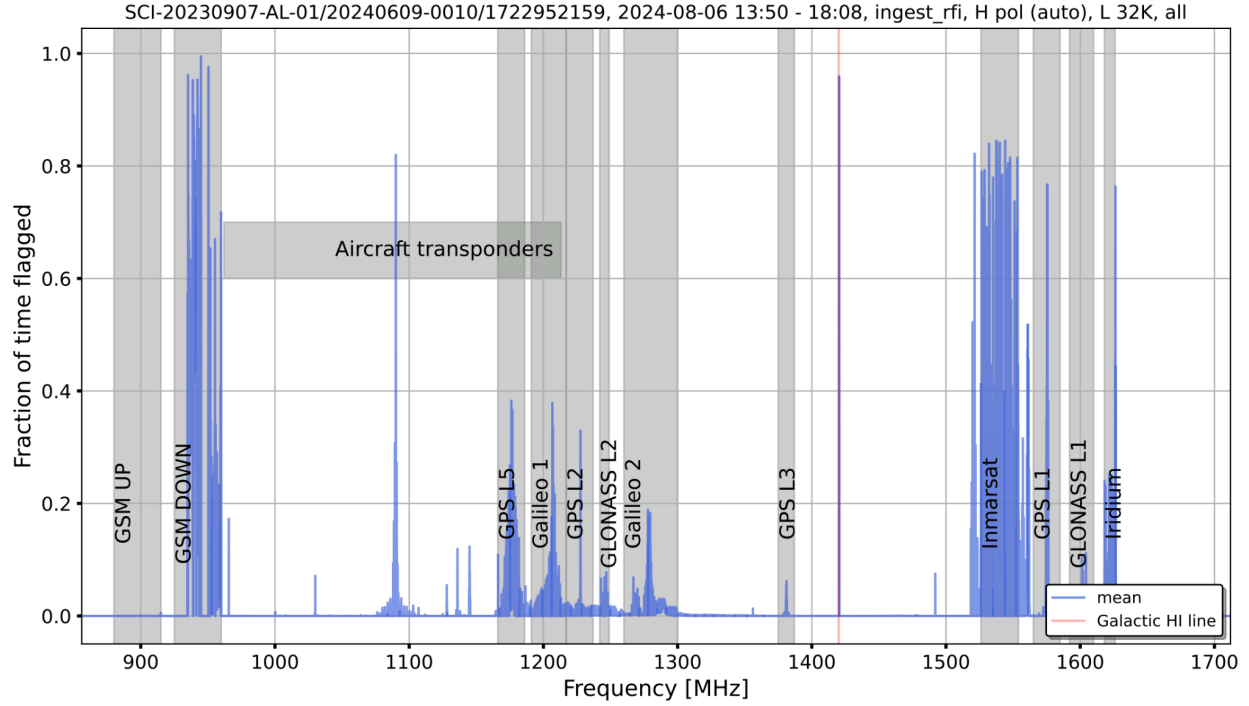


Figure 1: Mean of the autocorrelation flags of SB_ID:20240609-0010 dataset.

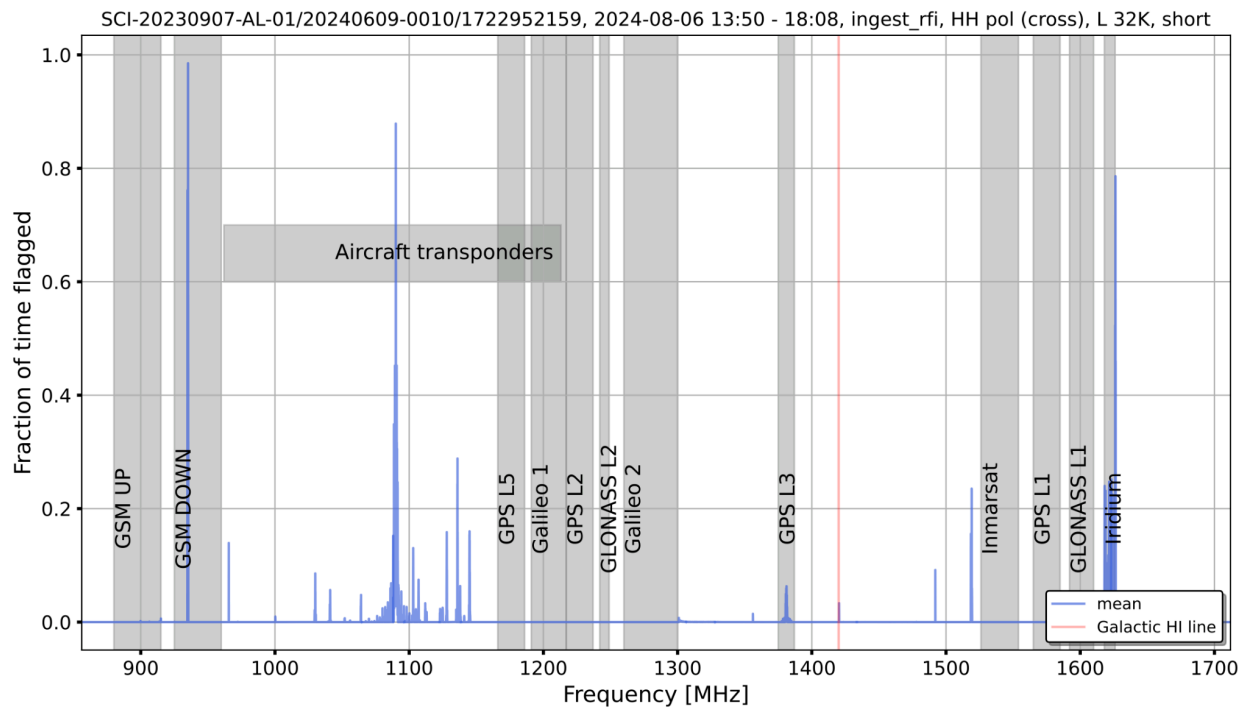


Figure 2: Mean of the cross correlation flags of the short baselines of SB_ID:20240609-0010 dataset.

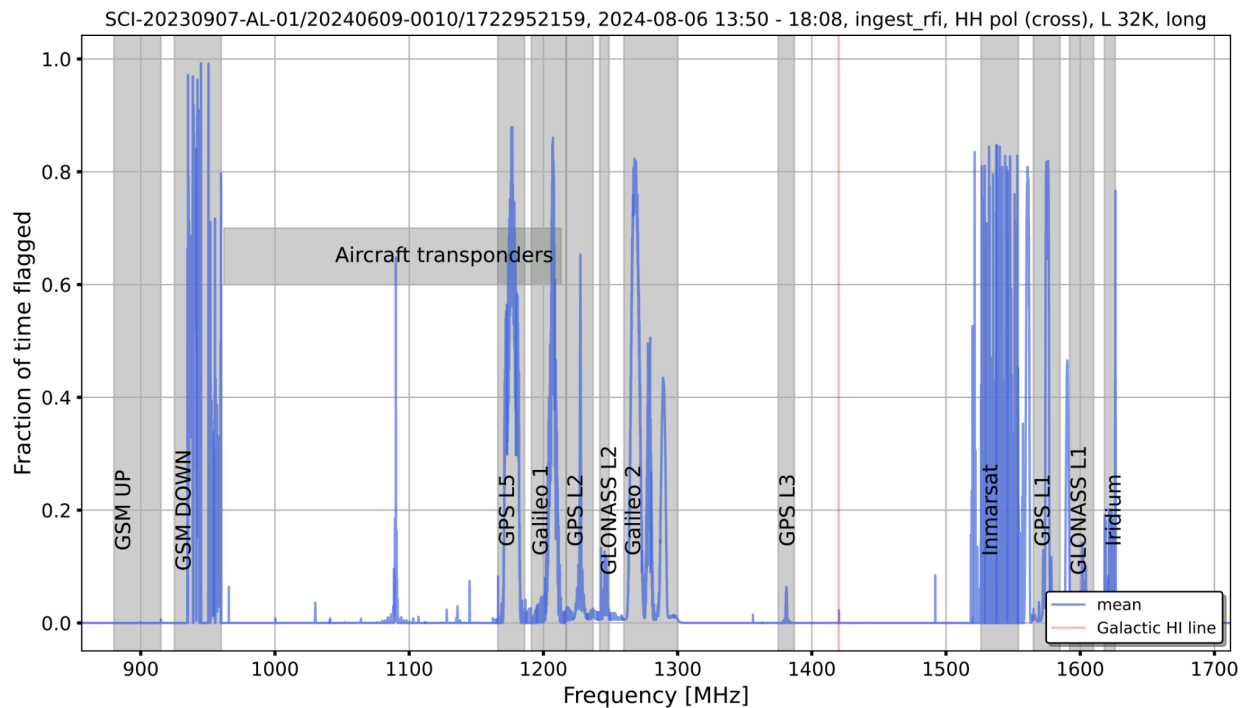


Figure 3: Mean of the cross correlation flags of long baselines of SB_ID:20240609-0010 dataset.

b. 2024, 13 August

The auto, short and long baselines cross correlations of the observation carried out on the 13th of August 2024 appear to be relatively clean (i.e. there are no unknown RFI emissions) as presented in Figures 4, 5 and 6. However, apart from the two emissions around 1500 MHz, there seems to be a weak noise emission close to the galactic HI line at 1420 MHz in the auto and short cross correlations flags (Figures 4 and 5).

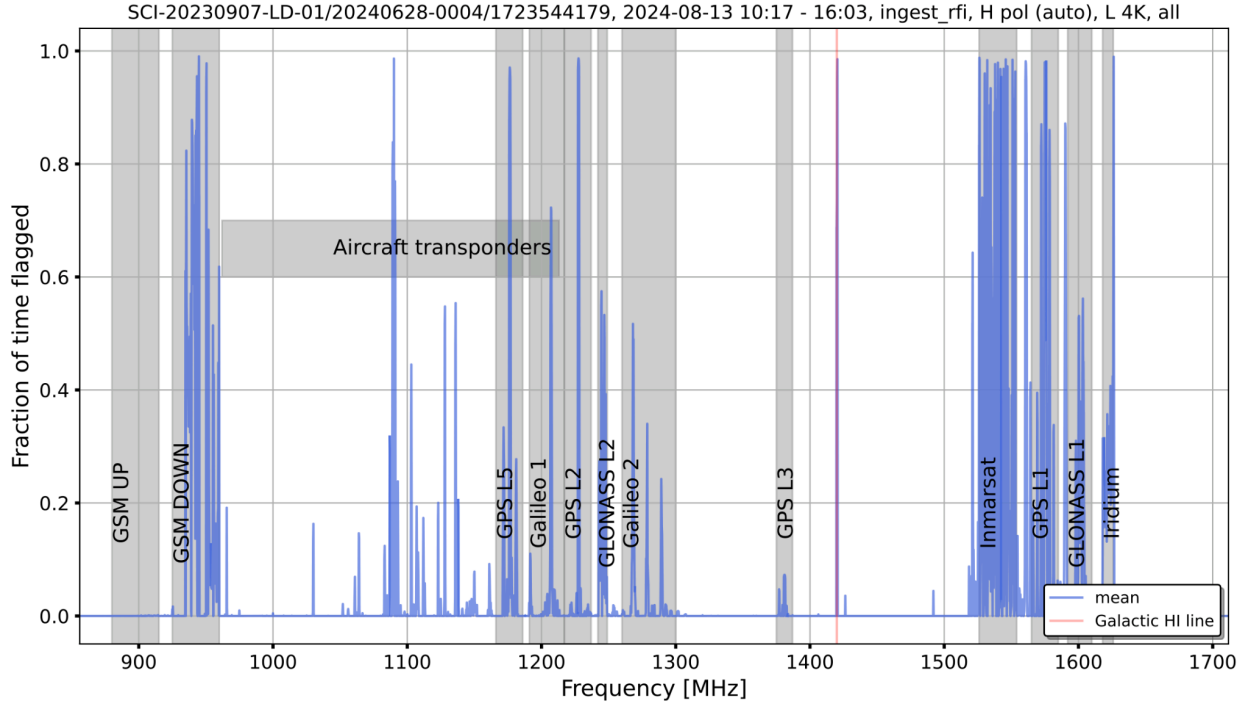


Figure 4: Mean of the autocorrelation flags of SB_ID:20240628-0004 dataset.

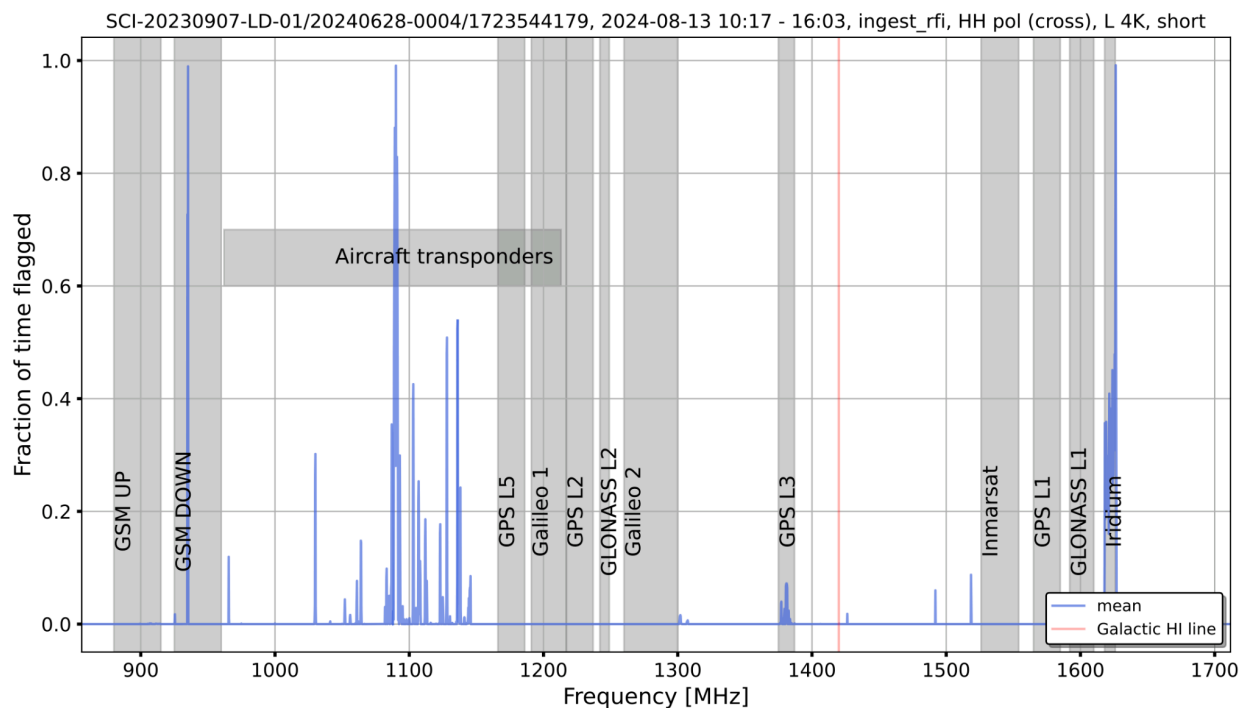


Figure 5: Mean of the cross correlation flags of the short baselines of SB_ID:20240628-0004 dataset.

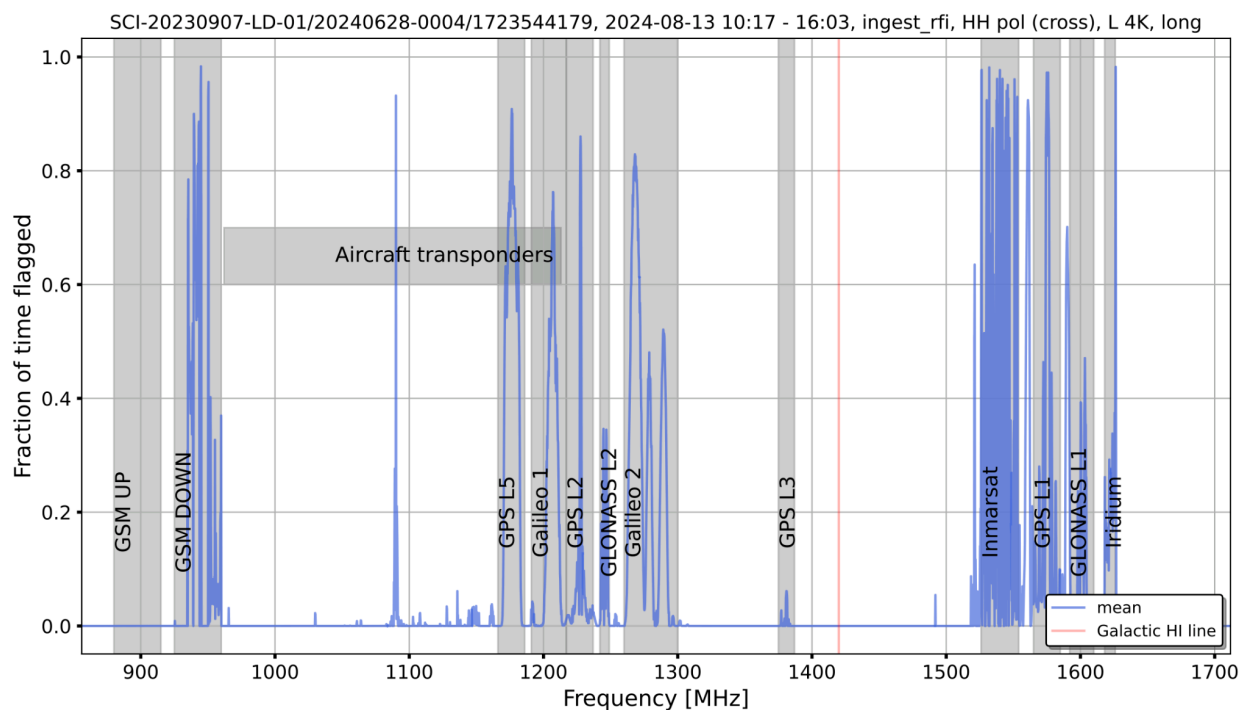


Figure 6: Mean of the cross correlation flags of long baselines of SB_ID:20240628-0004 dataset.

c. 2024, 17 August

The products from the observation on the 17th of August 2025 are relatively free of RFI as shown in Figures 7, 8 and 9.

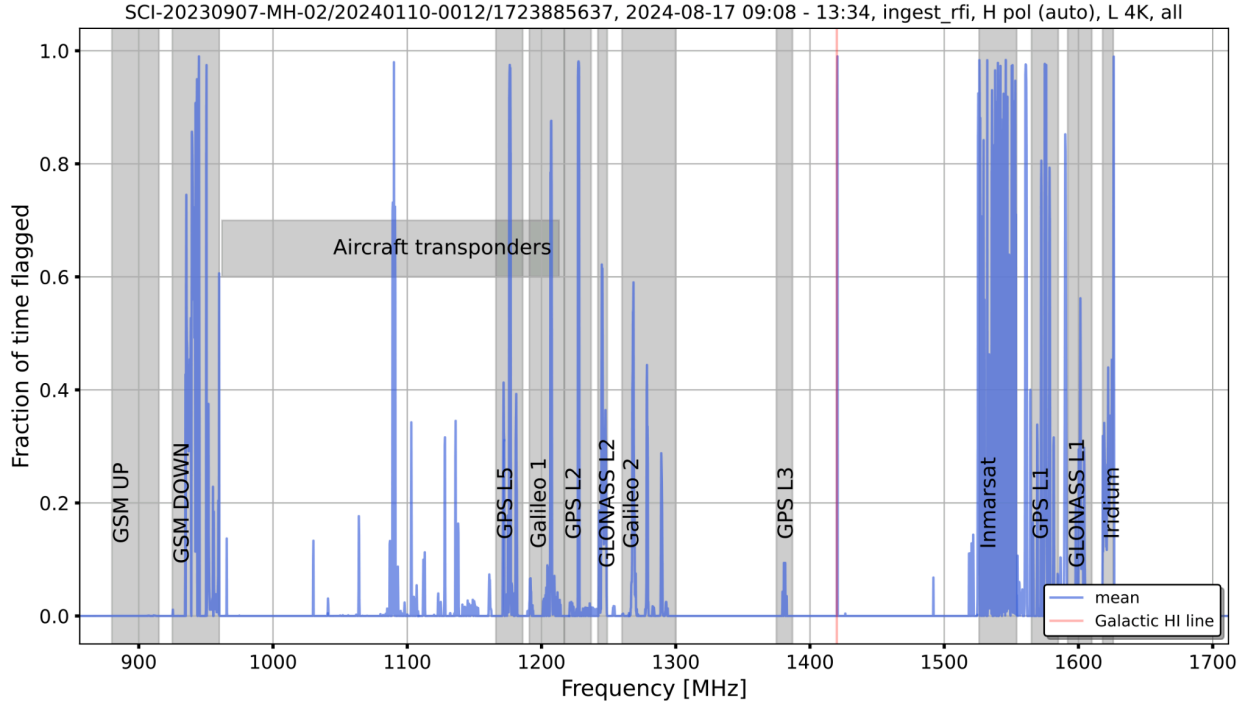


Figure 7: Mean of the autocorrelation flags of SB_ID:20240110-0012 dataset.

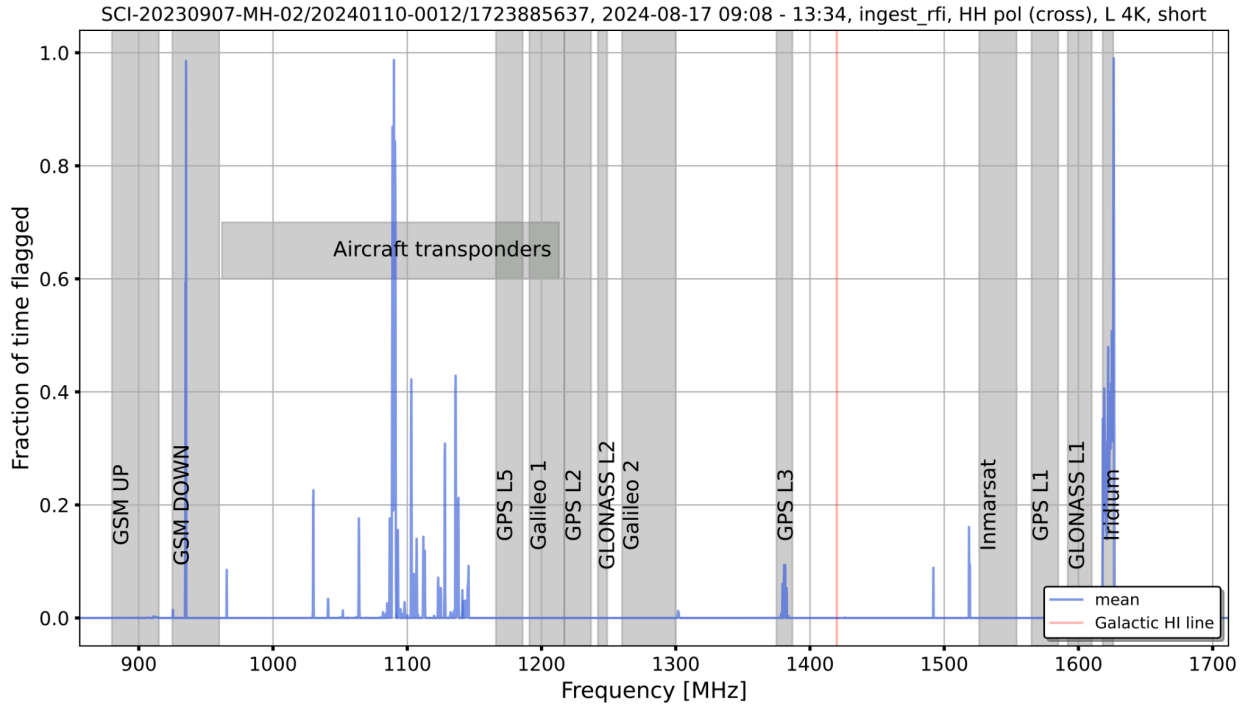
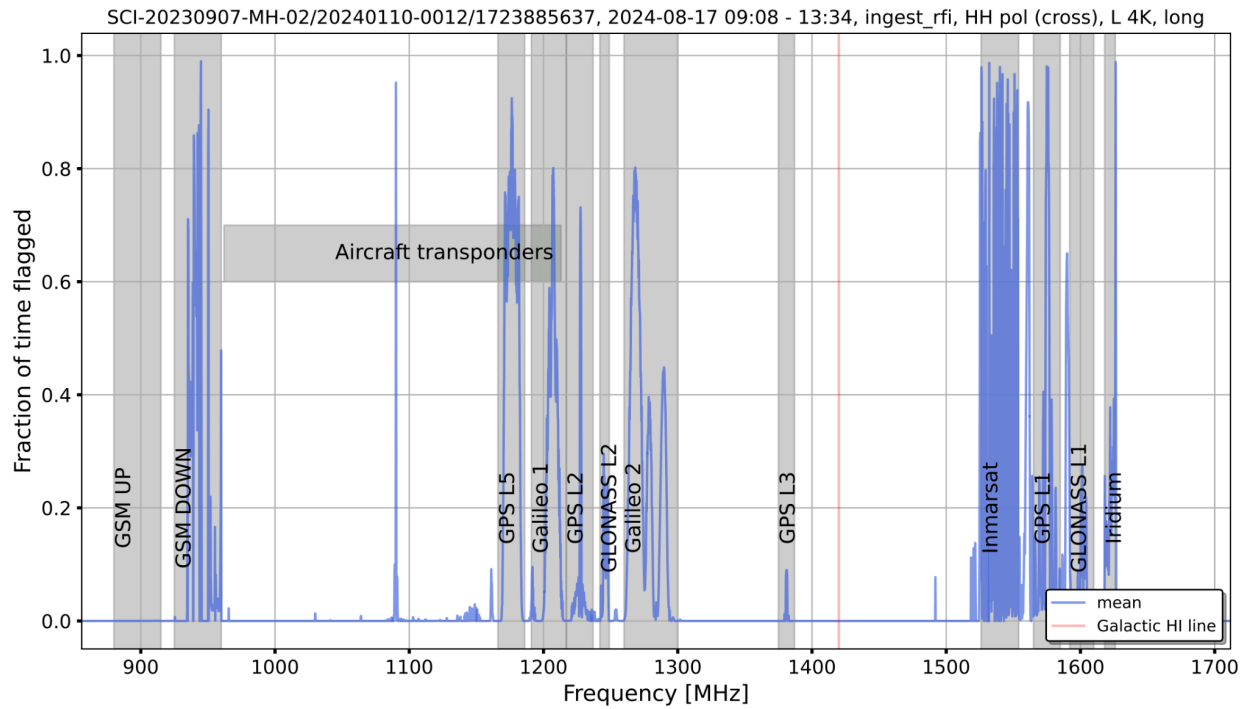


Figure 8: Mean of the cross correlation flags of the short baselines of SB_ID:20240110-0012 dataset.



d. 2024, 22 August

It appears that the auto and short baseline cross correlation products of the observation from 22nd of August 2025 exhibit varying degrees of contamination on a few channels. Apart from the contaminants around 1300 MHz and 1500 MHz, both autocorrelation and short baseline cross correlation flags are affected by extremely weak RFI in the quiet portion of the band (1300 - 1520 MHz).

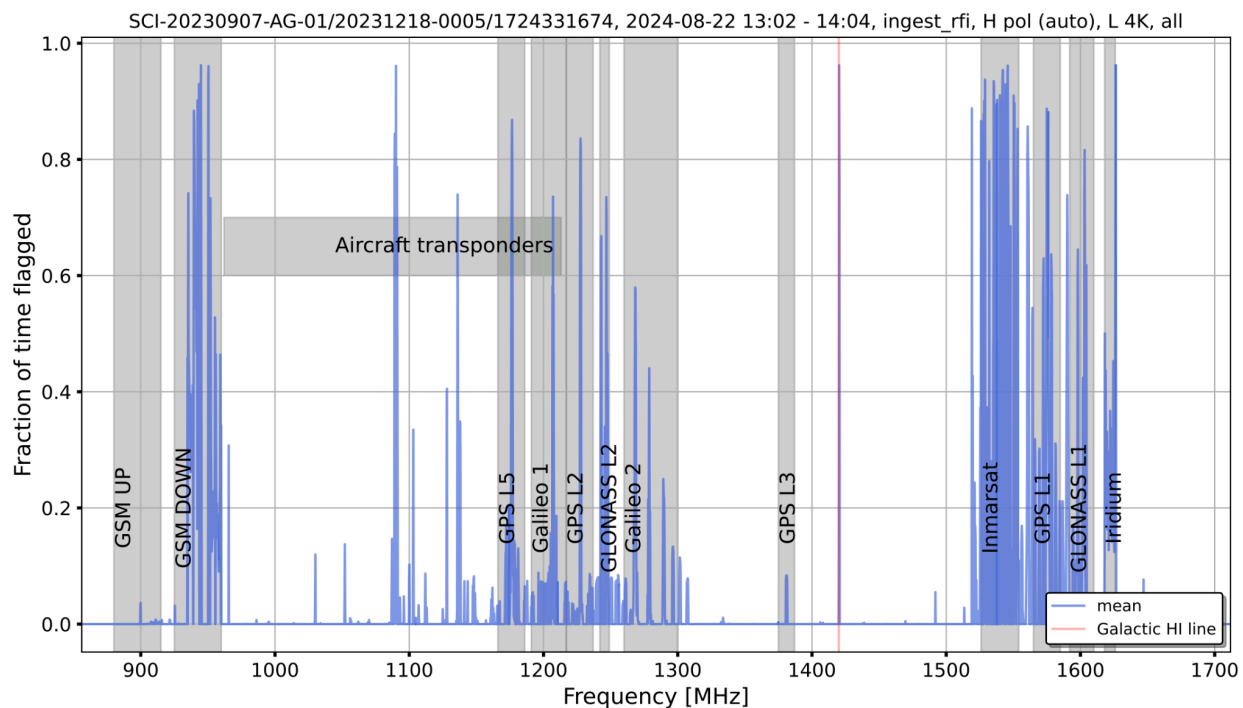


Figure 10: Mean of the autocorrelation flags of SB_ID:20231218-0005 dataset.

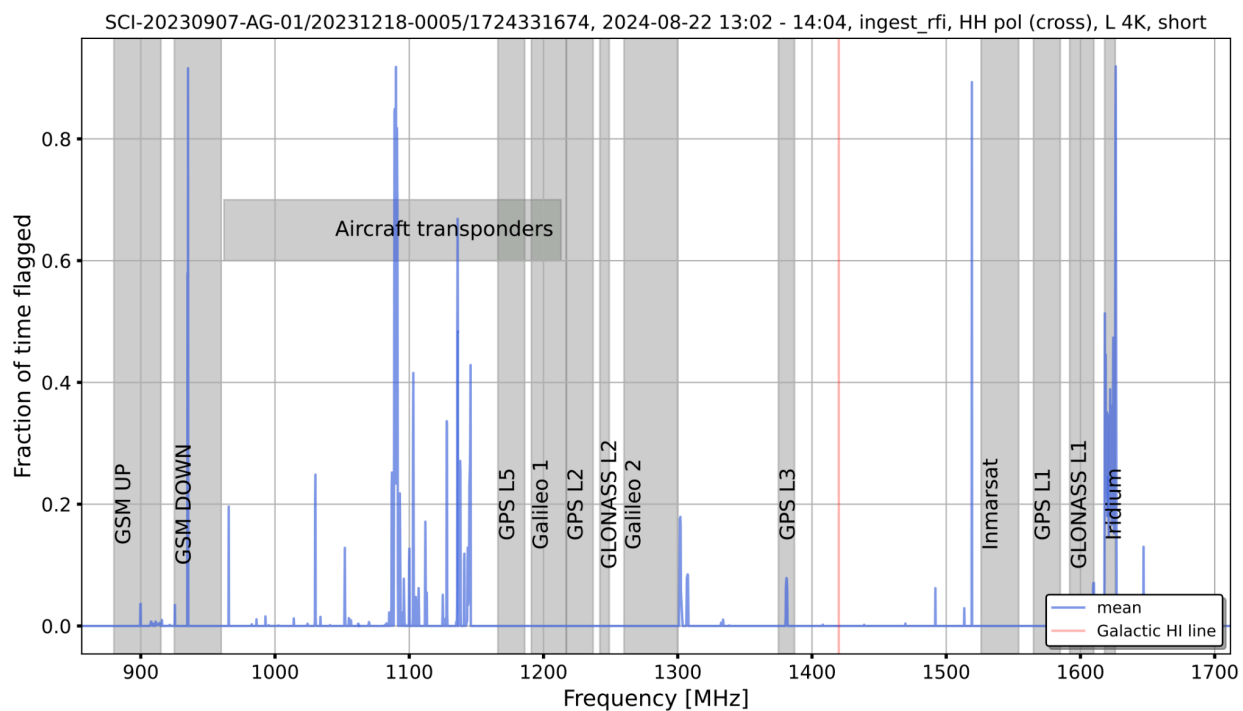


Figure 11: Mean of the cross correlation flags of the short baselines of SB_ID:20231218-0005 dataset.

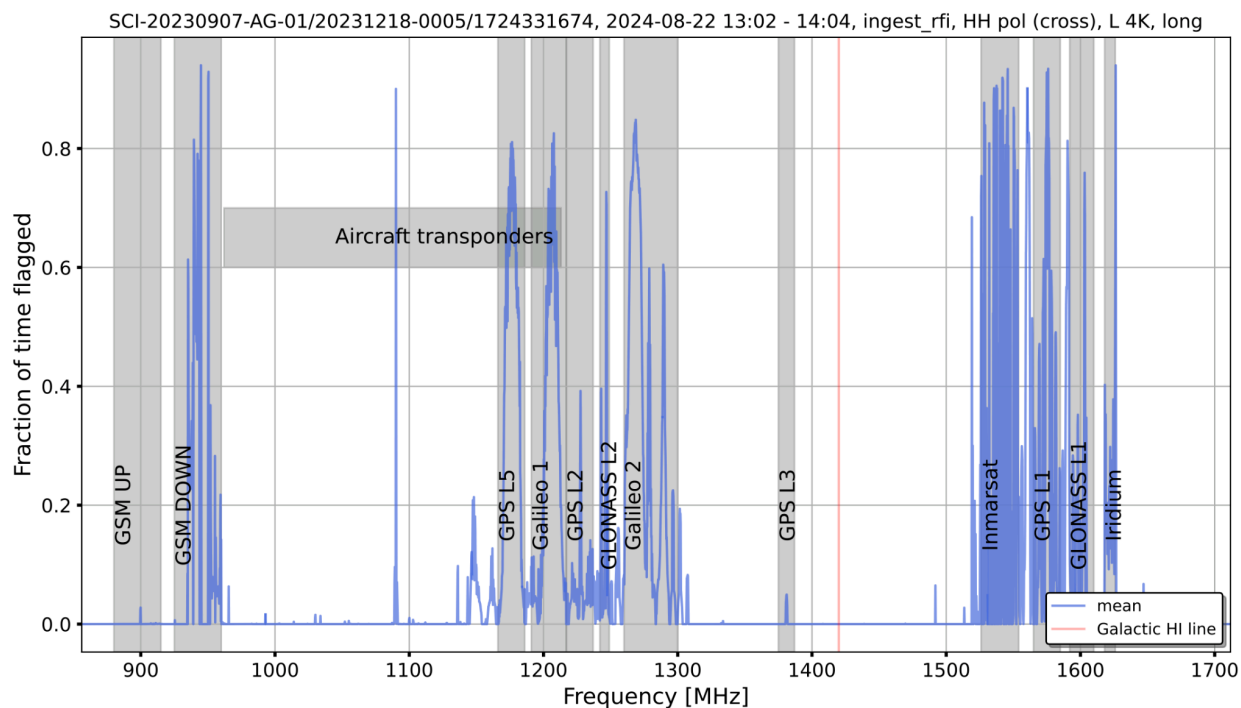


Figure 12: Mean of the cross correlation flags of the long baselines of SB_ID:20231218-0005 dataset.

e. 2024, 25 August

Overall, the data products from the observation on 25th of August 2025, as presented in Figures 13, 14 and 15, are clean.

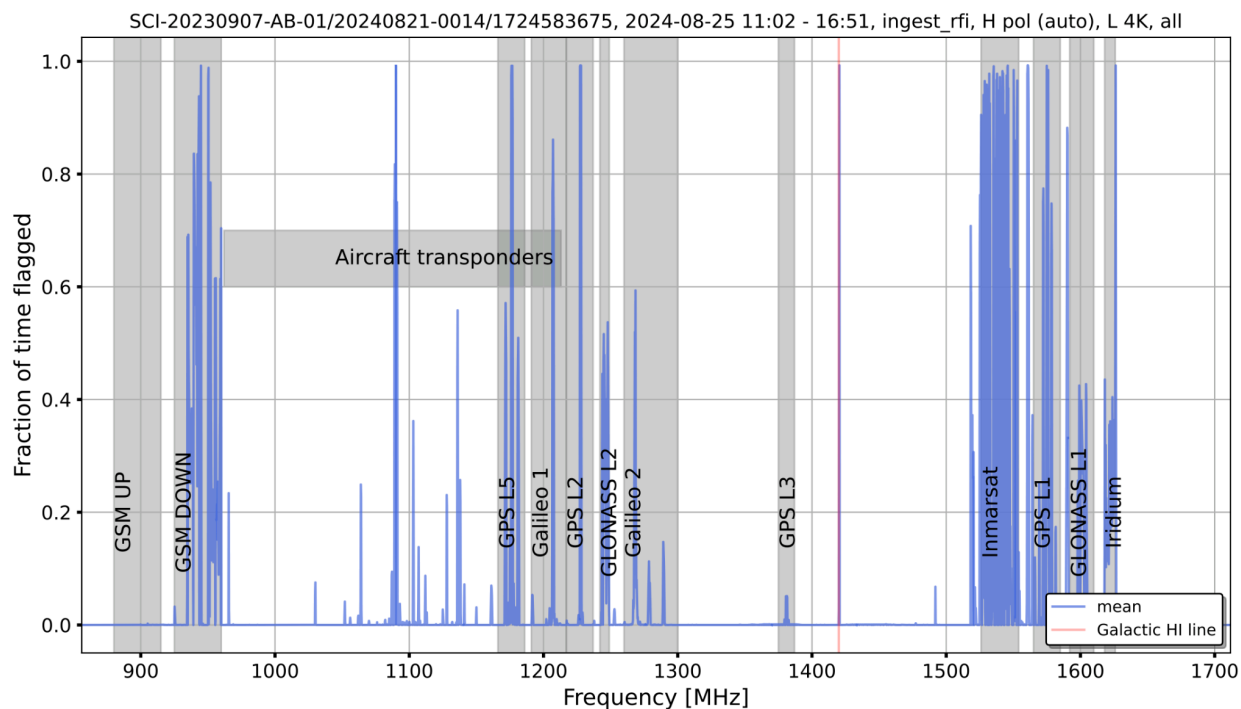


Figure 13: Mean of the cross correlation flags of the long baselines of SB_ID:20240821-0014 dataset.

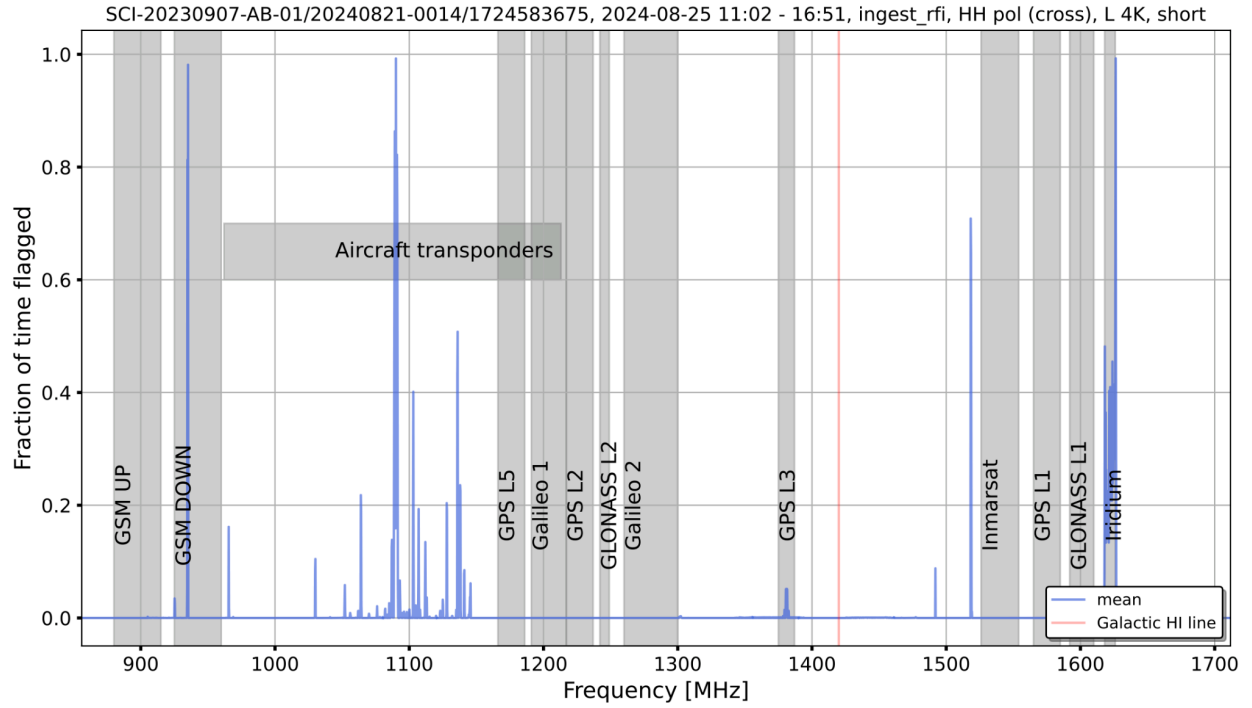


Figure 14: Mean of the cross correlation flags of the short baselines of SB_ID:20240821-0014 dataset.

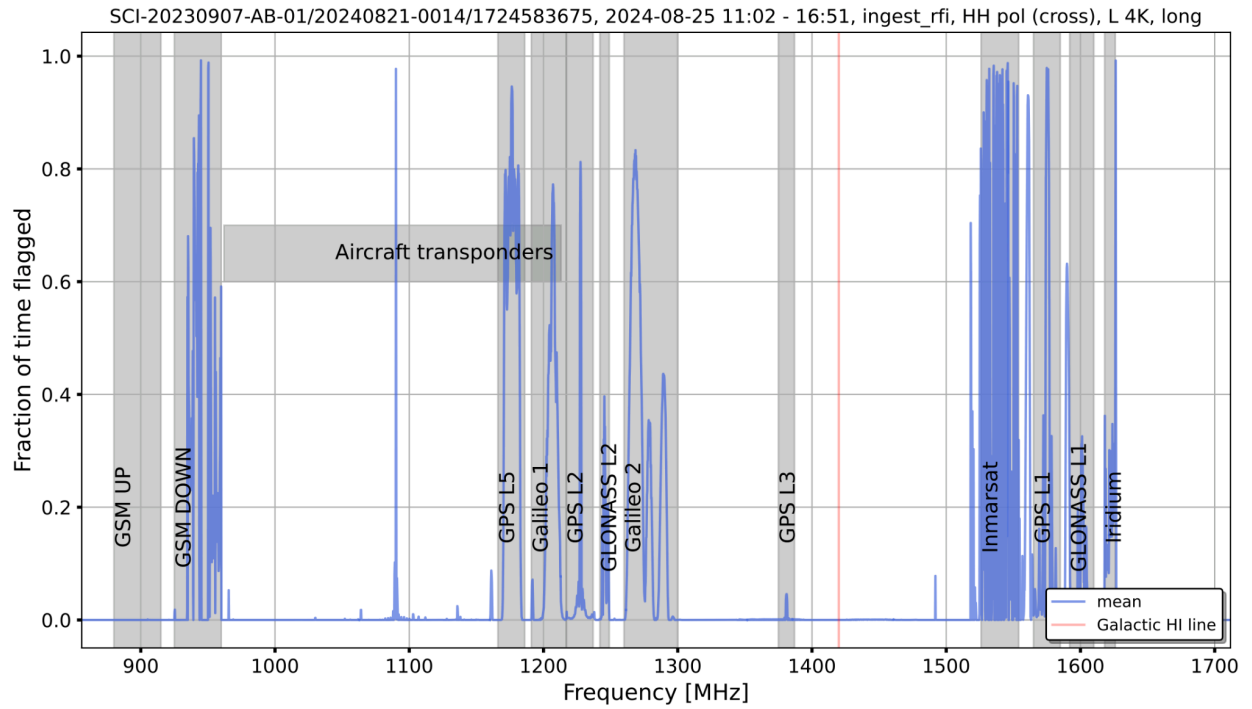


Figure 15: Mean of the cross correlation flags of the long baselines of SB_ID:20240821-0014 dataset.

4. RFI summary for November 2024 at UHF, HH cross polarization

Now we look at the overall RFI for the month of August by concatenating all science observations for the month. Here we select particularly 8s observations, observed at 4k or 32k channel mode. The monitoring and analysis of RFI are presented using the KATHPRFI pipeline [1], highlighting the impact of construction on RFI levels.

KATHPRFI algorithm

The `kathprfi` essentially processes two numpy arrays, `Master` and `Counter`, to update the number of RFI points and observations per voxel which refers to 5-D array of Time, Frequency, Baseline, Elevation and Azimuth. It iterates over chunks of frequency indices, baseline indices, and time indices. For each combination, the algorithm updates the `Master` array by the corresponding 'Good flags' value and the `Counter` array by 1. Good flags for a given observation are based on a flag type, e.g. if flag type is calibration RFI then target tags for that observation will be obtained from a complete calibration report. The output of this is an xarray dataset that contains arrays of these two arrays and the dimensions of interest as mentioned above.

Computation of marginal probability

The RFI probability is computed by using two arrays from the output Zarr file obtained from the `kathprfi` pipeline : 'Master' and 'Counter'. These arrays represent the number of RFI points and the total number of observations, respectively. The function sums these arrays along a specified dimension to get total counts and then calculates the probability as the ratio of these sums.

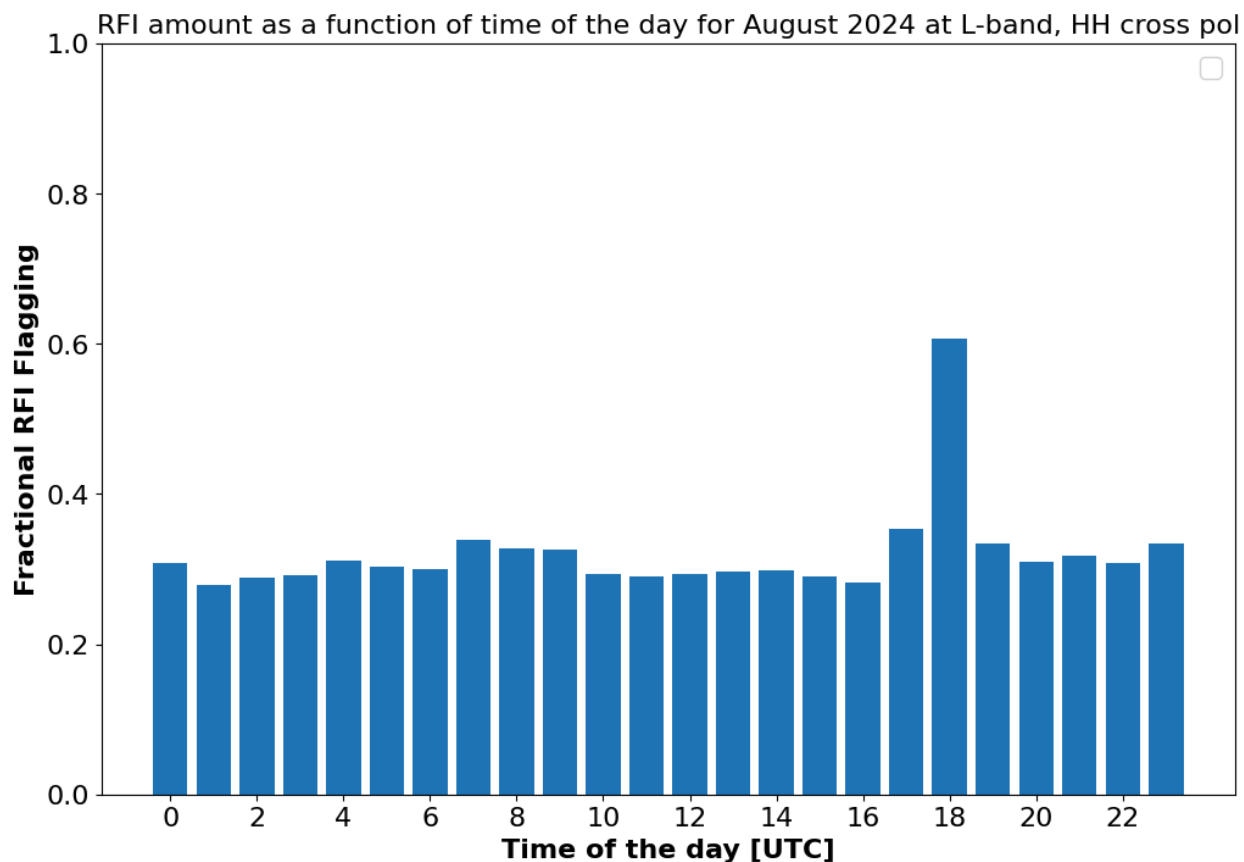


Figure 16: The above plot presents the fractional RFI flagging for August 2024 L-band horizontal cross pol observations, shown as a function of time of day in UTC.

Figure 16 illustrates the hourly behavior of RFI for November 2024, highlighting specific time periods affected by elevated RFI levels. The data reveals that RFI behavior is generally stable for L-band, with an average RFI percentage of 32% throughout the day. Notably, this is because there are more RFI sources for the frequency range of this band. RFI levels exceed the daily average during the hour 18:00 UTC. During this hour ~60% of RFI is observed. This suggests a period of heightened RFI prevalence. The causes for such elevated RFI levels may include specific construction activities on-site or interference related to a source under observation during that time. To better characterize these, a detailed analysis of the frequency spectra during these hours is necessary.

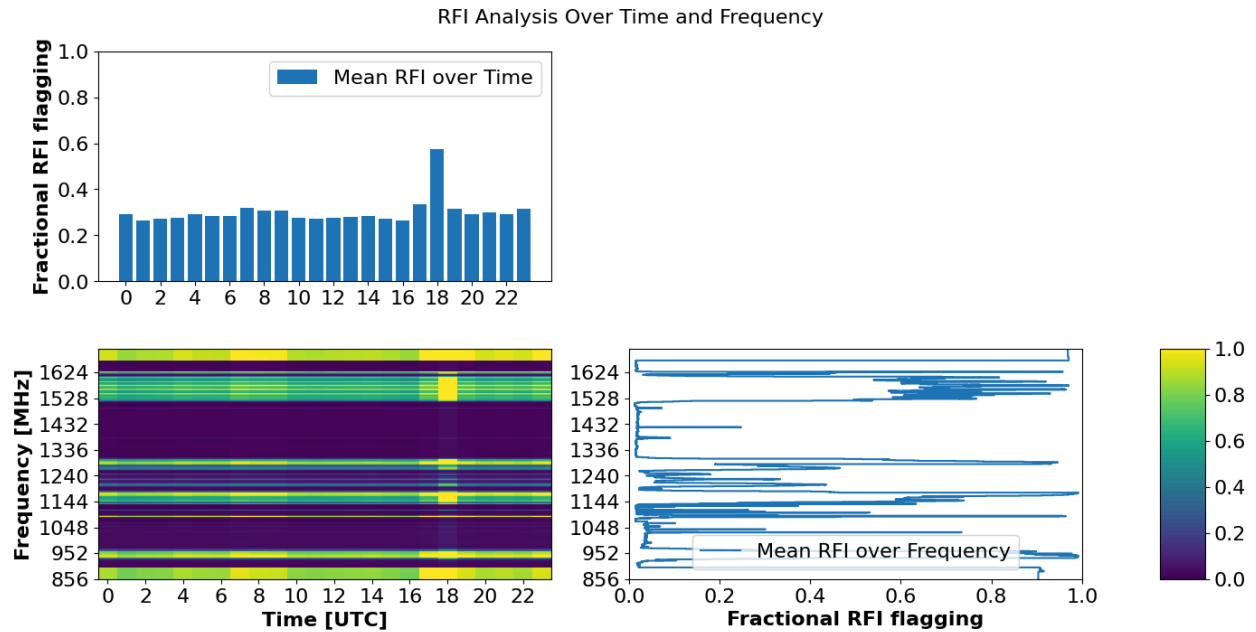


Figure 17: 2-D average RFI occupancy across time and frequency space.

Figure 17 presents a two-dimensional view of RFI patterns across time and frequency. The top 1D bar plot displays mean RFI levels over time, consistent with the trends shown in Figure 16. The central 2D heatmap provides a detailed visualization of RFI activity, with white regions indicating no observation at the hour. Intense RFI zones are represented by bright yellow or green patches, as per the color bar on the far right. Significant RFI is observed at all the known free-contaminated ranges for L-band. There is nothing unusual at the 18th hour across the band, what we can say is that more RFI was observed in that hour than any other hour of the day.